

Summary table of performance parameter comparison between Dowill Solvent-free Green Marine Coating G50 and general marine epoxy products

Application properties of DW ULV G50



No.	Test Item	Test Requirements	Test Result		Test Standard	Rmk.
			Universal Epoxy	DW ULV G50		
1	Condition in the container	Free of hard lumps after stirring and mixing, and in a uniform condition	Normal	Normal	Clause 6.1 of GB/T 1727-2021	
2	Color	The color of each coat shall be contrasting, and the topcoat shall be light-colored.	Oxide red, gray	Oxide red, gray	Visual	
3	Density, g/ml	1.30–1.60	1.42	1.52	GB/T 6750-2007	
4	Viscosity (KU)	≥80	78	90	GB/T 9269-2009	
5	Drying time	Surface dry, ≤4 h	2	4	GB/T 1728-2020	
		Through dry, ≤24 h	24	24		
6	Pot life	≥1 h	4	1	GB/T 31416-2015	
7	Min. interval	天	1	1	Ambient temperature 20°C	
8	Max.interval	天	7	7	Ambient temperature 20°C	
9	Min.immersion time	天	7	7	Ambient temperature 20°C	
10	Application method		Airless spraying, air spraying, roller , brush	Airless spraying, air spraying, roller , brush		
11	Application equipment		Changjiang/GRACO	Changjiang/GRACO		
12	Nozzle tip		17B25	17B25		
13	Input pressure	kg	5	6		
14	Pump ratio		60/1	65/1		
15	Nozzle pressure	Mpa	30	39	The pipeline has pressure degreasing.	
16	Application Temperature	°C	10-40	10-40		
17	Application Humidity	%	≤85	≤85		
18	Storage stability	Pass after 30 days at 50°C	Pass	Pass	GB/T 6822-2024 Section 6.4.5	
19	Applicability	Smooth application without obstruction	Normal	Normal	GB/T 6822-2024 Section 6.4.5	

Environmental Performance of DW ULV G50

No.	Test Item	Test Requirements	Test Result		Test Standard	Rmk.
			Universal Epoxy	DW ULV G50		
1	Non-volatile	≥70%	82	96	GB/T 1725-2007	
2	SVR	≥75%	80	95	GB/T 9272-2007	
3	Theoretical coverage	m²/L/160µm	5.00	5.94	Theoretical value	
4	Application SVR	%	72	95	GB/T 9272-2007	
5	Flash point (°C)	Closed cup method	≥30	≥63	GB/T 261-2021	
6	VOCs	≤395g/L	160	60	GB/T 23985-2009	
7	Application VOCs	g/L	170	60	GB/T 23985-2009	
8	VOCs/200 µ	g/m²	47	13	GB/T 23985-2009	
9	Total content of toluene and xylene(%)	≤35%	12	≤1	GB/T 23990-2009	
10	Benzene content (%)	≤0.3	Not detected	Not detected	GB/T 23990-2009	
11	Total glycol ether and ether ester content (%)	≤1	Not detected	Not detected	GB/T 23986-2009	
12	Lead content (mg/kg)	≤1000	Not detected	Not detected	GB/T 30647-2014	
13	Cadmium content (mg/kg)	≤100	Not detected	Not detected	GB/T 30647-2014	
14	Hexavalent chromium content (mg/kg)	≤1000	Not detected	Not detected	GB/T 30647-2014	
15	Mercury content (mg/kg)	≤1000	Not detected	Not detected	GB/T 30647-2014	
16	Asbestos	None	Not detected	Not detected	ISO 22262-1 and NIOSH 9000	

Dowill High-Solid, Low-Viscosity, Solvent-Free Green Marine Coatings

Compared with traditional solvent-based marine coatings, our products feature superior anti-corrosion performance and 30–50% higher coating application efficiency with a smooth coating surface. They are widely applicable to ballast tanks, cargo holds, decks, ship bottoms and hulls. VOC emissions from full-vessel coating can be reduced by more than 80%, eliminating the need for VOC collection and incineration treatment. Carbon emissions can be cut by over 95%, helping shipyards save on fixed-asset investment and daily energy consumption.

The products are especially suitable for potable water tanks, cargo holds, engine rooms and crew quarters, particularly passenger cabins and recreational areas on cruise ships. They completely eliminate long-term residues of toxic substances such as VOCs and benzene carcinogens, as well as irritating odors from solvent-based coatings in enclosed ship spaces, protecting the health of crew members and passengers.

Solvent-Free Coating

DW ULV G50

Product Description	A solvent-free coating based on modified-epoxy resin of organic-inorganic. Complies with the VOCs limit standards of *GB/T 38597-2020 Technical Requirements for Low Volatile Organic Compound Content Coatings Products*, and is exempt from regulations restricting fugitive emissions during coating application.																													
Recommended Use	As a protective primer-finish coat can be used in ballast tank or similar location. As a primer for in-water service.																													
Complied Standards	Complies with GB/T6823-2008																													
Features	Ultra high-solid and environmentally friendly Excellent surface-tolerant properties Excellent anticorrosive performance Excellent resistance to impact and abrasion. Excellent high-build applicability Highly suitable recommended for drinking water tanks																													
Physical Properties																														
Color	White and Grey																													
Volume solids	96%																													
VOC	16g/L, accord with EU Dircrtive 2004/42/EC, subcategory j																													
Flash Point	Greater than 63°C																													
Film thickness	Dry: 100-300µm/coat Wet: 104-313 µm/coat																													
Theoretical Coverage	4.8 m2/L/200 µm																													
Surface Preparation	Practical coverage depending on surface conditions, environment, applications, etc. Remove oil and grease with suitable detergent or solvent(SSPC-SP 1). Remove rust, mill-scale and other loose material completely by abrasive blasting(ISO Sa2.5). All surface to be coated should completely clean, dry and free from contamination.																													
Application	5°C - 35°C																													
Application Temperature																														
Mixing	Combine Base with Hardener, mix thoroughly with power agitator																													
Mixing Ratio	Base/Hardener = 3.5/1(by Volume)																													
Pot Life	<table><tr><td>10°C</td><td>20°C</td><td>30°C</td></tr><tr><td>5hrs</td><td>2.5hrs</td><td>1.5hrs</td></tr></table>			10°C	20°C	30°C	5hrs	2.5hrs	1.5hrs																					
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Application Methods	Airless spray, air spray, brush and roller Airless spray Nozzle tip 0.42mm - 0.60mm Nozzle 15-21Mpa																													
Thinner	—																													
Drying time & Overcoating interval	<table><tr><td>Temperature</td><td>5°C</td><td>10°C</td><td>20°C</td><td>30°C</td></tr><tr><td>Surface dry</td><td>5hrs</td><td>5hrs</td><td>3hrs</td><td>2hrs</td></tr><tr><td>Dry to touch</td><td>24hrs</td><td>18hrs</td><td>12hrs</td><td>8hrs</td></tr><tr><td>Min.interval</td><td>48hrs</td><td>24hrs</td><td>16hrs</td><td>8hrs</td></tr><tr><td>Max.interval</td><td>30days</td><td>25days</td><td>15days</td><td>7days</td></tr></table>					Temperature	5°C	10°C	20°C	30°C	Surface dry	5hrs	5hrs	3hrs	2hrs	Dry to touch	24hrs	18hrs	12hrs	8hrs	Min.interval	48hrs	24hrs	16hrs	8hrs	Max.interval	30days	25days	15days	7days
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Min.interval	48hrs	24hrs	16hrs	8hrs																										
Max.interval	30days	25days	15days	7days																										
Applied Substrate	Bare steel																													
Typical topcoat	DW ULV G50 etc.																													
Storage	Store in dry, cool condition and away from sources of heat and ignition. Containers must be kept tightly closed.																													
Shelf life	24 months from date of production (25°C)																													

Hard Antifouling Coating G60

DWAFMARINE G60

Product Description	An anti-fouling solvent-free hard paint based on silicon modified-epoxy resin of organic-inorganic. Complies with the VOCs limit standards of *GB/T 38597-2020 Technical Requirements for Low Volatile Organic Compound Content Coatings Products*, and is exempt from regulations restricting fugitive emissions during coating application.				
Recommended Use	Can be used in bottoms & below water of steel structure with underwater robot as new construction or maintenance situations for ship industry etc.				
Complied Standards	Complies with GB/T6822-2024				
Features	Ultra-high-solid content, environmentally beneficial Ultra-smooth post-docking, biocide-free Hard, cleanable surface with superior impact resistance Prevents fouling attachment and settlement Zero leaching of biocides or microplastics				
Physical Properties					
Color	Oxide red and Grey				
Volume solids	95%				
VOC	70g/L, accord with EU Dircrtive 2004/42/EC, subcategory j				
Flash Point	Greater than 63°C				
Film thickness	Dry: 100-300µm/coat Wet: 105-316 µm/coat				
Theoretical Coverage	4.8 m2/L/200 µm				
Surface Preparation	Practical coverage depending on surface conditions, environment, applications, etc. Remove oil and grease with suitable detergent or solvent(SSPC-SP 1). Remove rust, mill-scale and other loose material completely by abrasive blasting(ISO Sa2.5). All surface to be coated should completely clean, dry and free from contamination.				
Application	5°C - 35°C				
Application Temperature					
Mixing	Combine Base with Hardener, mix thoroughly with power agitator				
Mixing Ratio	Base/Hardener = 2.5/1(by Volume)				
Pot Life	10°C	20°C	30°C		
	5hrs	2.5hrs	1.5hrs		
Application Methods	Airless spray, air spray, brush and roller Airless spray Nozzle tip 0.42mm - 0.60mm Nozzle 15-21Mpa				
Thinner	—				
Drying time & Overcoating interval	Temperature	5°C	10°C	20°C	30°C
	Surface dry	5hrs	5hrs	3hrs	2hrs
	Dry to touch	24hrs	18hrs	12hrs	8hrs
	Min.interval	48hrs	24hrs	16hrs	8hrs
	Max.interval	30days	25days	15days	7days
Applied Substrate	G50 or bare steel				
Storage	Store in dry, cool condition and away from sources of heat and ignition. Containers must be kept tightly closed.				
Shelf life	24 months from date of production (25°C)				

Summary table of performance parameter comparison between Dowill Solvent-free Green Marine Coating G50 and general marine epoxy products

Physical and chemical properties of DW ULV G50

No.	Test Item	Test Requirements	Test Result		Test Standard	Rmk.
			Universal Epoxy	DW ULV G50		
1	Color and appearance	Normal	Normal	Normal	GB/T 6822-2024, Clause 6.4.6	
2	Gloss		8-15	40-60	GB 1743-1979	
3	Adhesion	≥5.0MPa	7-12	8-15	GB/T 5210-2016	
4	Immersion resistance	Pass after 140 days	Pass	Pass	GB/T 10834-2008	
5	Blister Resistance	No blistering after 2 cycles	No blistering	No blistering	GB/T 10834-2008	
6	Cathodic disbondment	≤8mm	3-4	3-4	GB/T 7790-2008	
7	Salt spray test	After 1,000 hours, the paint film should show no blistering, no peeling, and no rusting	After 1,000 hours, the paint film shows no blistering, no peeling, and no rusting	After 1200 hours, the paint film shows no blistering, no peeling, and no rusting	GB/T 1771-2007	
8	Salt water test	At 40±2°C for 336 hours, the paint film should show no blistering, cracking, peeling, wrinkling, or rust	After 336 hours, the paint film shows no blistering, cracking, peeling, wrinkling, or rust	After 600 hours, the paint film shows no blistering, cracking, peeling, wrinkling, or rust	GB/T 10834-2008	
9	Oil resistance	After 21 days in gasoline and diesel, the paint film should show no blistering, no peeling, and no softening	After 21 days in gasoline and diesel, the paint film shows no blistering, no peeling, and no softening	After 30 days in gasoline and diesel, the paint film shows no blistering, no peeling, and no softening	GB/T 9274-1998	
10	Acid resistance (5% H ₂ SO ₄)	After 480 hours, the paint film should show no peeling and no blistering	The paint film shows no peeling and no blistering	The paint film shows no peeling and no blistering	GB/T 9274-1998	
11	Alkali resistance (5% NaOH)	After 480 hours, the paint film should show no peeling and no blistering	The paint film shows no peeling and no blistering	The paint film shows no peeling and no blistering	GB/T 9274-1998	
12	Abrasion resistance	CS-10, 500 g, 500 cycles, weight loss not exceeding 100 mg	50	25	GB/T 1768-2006	
13	Hardnes		2H	3H	GB/T 6739-2006	
14	Flexibility	Not greater than 3 mm	2	2	GB/T 1731-2020	
15	Impact resistance	Not less than 40 kg (specified as subject to agreement in the GB standard)	50	50	GB/T 5210-2006	
16	Hygienic requirements	Grain hold certification.	Pass	Pass	GB 4806.10-2016	
17	Diesel resistance	After half a year, the paint film should show no blistering, softening, peeling, or rust	After 6 months, the paint film shows no blistering, softening, peeling, or rust	After 8 months, the paint film shows no blistering, softening, peeling, or rust	GB/T 9274-1998	
18	Simulated ballast tank condition test	Pass	Pass	Pass	GB/T 6823-2008, Annex A	
19	Condensation chamber test	Pass	Pass	Pass	GB/T 6823-2008, Annex B	
20	Ballast tank crossover test	Pass	Pass	Pass	GB/T 6823-2008 IMO MSC.215(82) Performance Standard for Protective Coatings for Dedicated Seawater Ballast Tanks in All Types of Ships and Double-side Skin Spaces of Bulk Carriers	
21	Gas-tight chamber test	90 days	Pass	Pass	MSC.288(87) and GB/T 31820-2015	
22	Immersion test	180 days	Pass	Pass	GB/T 31820-2015, Annex B	